MILITARY SPECIFICATION

MINE, CHEMICAL AGENT, VX, M23

1. SCOPE

1.1 This specification covers a land mine filled with chemical agent. VX .

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-D-51083 - Detector Ticket, Anticholinesterase Agent,
MIL-B-51189 - Substrate Solution and Dispenser.
MIL-C-52078 - Cap-Plug, Cap and Plug, Protective, Plastic Dust and Moisture Seal.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAMINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL37-1-13 - Nine, Chemical Agent, VX, M23.

FSC 1345

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:28Z
Check the source to verify that this is the current version before use.

PICATINNY ARSENAL

F7548391 - Ammunition Container, Metal, for Activator, Ml; Activator Practice, ML, and Activator, M2, Assembly, Details, Packing and Marking.
F7549159 - Container Ammunition, Metal, for Fuze, Mine, AT, M603. Assembly, Details, Packing and Marking.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting office.)

3. REQUIREMENTS

3.1 Materials and components. The supplier shall ascertain prior to their use that all materials and components have met their respective specifications.

3.2 Assembly.

3.2.1 Filling and closing. The mine shall be filled and closed (see 6.2 and 6.3) as shown on Drawing D37-1-13.

3.2.2 Explosive components. The 1148 initiator, Drawing C37-1-64, shall be placed in the side burster tube so that the below flush charge end (if any) is toward the outside periphery of the mine and secured in place with the snap ring as shown on D37-1-13. The side burster tube opening shall be closed with cap plug conforming to MIL-C-52078. The M38 burster, Drawing C37-1-63, shall be placed in conical cavity in the bottom of the mine. The opening shall be closed as shown on Drawing D37-1-13. The M120 booster shall be secured in the M38 as shown on Drawing D37-1-13 prior to inserting The burster into the mine.

3.3 Agent contamination.

3.3.1 Mine. The mine shall be free from agent contamination when tested in accordance with 4.2.4.1(a).

3.3.2 Closure leakage and contamination. The closure shall be free from leakage and contamination when tested in accordance with 4.2.4.1(b).

3.4 Helium leakage. The mine closure shall not leak helium at a rate greater than 1×10^{-6} atmosphere cubic centimeter per second when tested in accordance with 4.2.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:28Z
Check the source to verify that this is the current version before use.

3.5 Filling level The filling level in the mine shall be within applicable limits shown in table I when examined in accordance with 4.2.4.3.

Table I. Filling level

Temperature range °F	Void (Distance from datum filling line to liquid surface.)	
	Minimum	Maximum
Between 40 - 50	2.12	2.64
50 - 60	2.07	2.59
60 - 70	2.01	2.55
70 - 80	1.96	2.50
80 - 90	1.90	2.46
90 - 100	1.85	2.41

Calibration factors

9 percent void at 40° F in a minimum mine
Expansion factor VX + 0.00046 ml/O F - ml agent

3.6 Workmansi The mine shall be free from damage which could adversely affect subsequent assembly, handling, storage, safety and functioning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and 3.2 for which specific inspection has not been provided in this specification have been satisfied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:28Z
Check the source to verify that this is the current version before use.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the mines of one interfix number, filled an assembled by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions. However, no more than one lot of agent, fuzes, activators, side initiators, bursters and boosters shall be represented in any one lot of filled mines.

4.2.2 Sampling.

4.2.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For filling level and closure contamination. Sampling shall be conducted in accordance with MIL-STD-1235 (see 6.9).

4.2.3 Inspection procedure.

4.2.3.1 For examination and nondestructive tests. Sample mines and the level A preparation for delivery shall be examined in accordance with the classification of defects and with M1L-STD-105.

4.2.3.2 Decontamination process. Fifty mines from one shift produced under regular production conditions shall be tested in accordance with 4.2.4.1(a) to Qualify the decontamination procedure or whenever the decontamination procedure-is modified.

4.2.3.3 For filling level and closure contamination. The frequency of tests for filling level and closure contamination shall be in accordance with CSP 1, MIL-STD-1235 using an AQL of 0.25 percent defective for filling and 1.0 percent defective for closure contamination.

4.2.3.4 For criterial defects. Each mine in the lot shall be inspected for the Criterial characteristics listed in the classification of defects.

Source: https://assist.dla.mil -- Downloaded: 202640-22T07:28Z
Check the source to verify that this is the currentversion before use.

4.2.3.5 Classification of defects.

(a) Mine, chemical agent, VX, M23 - assembly (Dwg D37-1-13).

Categories	Defects	Acceptance standards
Critical:		
1	Helium Leakage	4.2.4.2
2	Agent leakage	
3	Color code incorrect	
4	Arming plug warning marking incorrect	
Major:	AQL 0.65 percent defective	
101	Torque on fuze well incorrect	
	AQL 1.0 percent defective	
102	Marking missing or incorrect	
103	Fuze retainer spring missing	
104	M120 booster incorrectly assembled to M38 burster	
105	M120 booster retaining ring damaged or missing	
106	M38 burster missing	
107	M38 burster incorrectly assembled to mine	
108	M38 burster damaged	
109	Fuze well gasket missing (O-ring) (Dwg B37-1-18)	
110	Felt gasket missing (Dwg A37-1-19)	
111	Burster tube incorrectly assembled	
112	M48 initiator damaged or snap ring missing	
113	Mine body damaged (dents, cuts, or distortion due to internal pressure)	
114	Internal thread of burster tube damaged	
115	Adapter thread damaged (Dwg 37-1-42)	
116	Fuze well thread damaged	
117	M120 Booster missing	
118	M120 Booster damaged	
119	M48 initiator missing	

Source: https://assist.dla.mil -- Downloaded: 202610-22T07:28Z
Check the source to verify that this is the current version before use.

4.2.3.5 Classification of defects (continued).

(b) Preparation for delivery, intermediate Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	M603 fuze missing
102	M1 activator missing
103	Instruction card missing
104	Internal cushioning and blocking within steel drum incorrect
105	Steel drum closure incorrect
106	Steel drum marking incorrect
107	Steel drum marking illegible

(c) Preparation for delivery, palletized load (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Internal crate blocking incorrect or missing
102	Crate incorrectly assembled
103	Crate damaged
104	Crate marking incorrect or illegible

4.2.4 Tests.

4.2.4.1 Agent contamination (see 6.4). The supplier shall submit to the Government his procedure for decontamination of filled and closed mines, including his methods of controlling his process, and shall obtain written approval from the Government prior to initiating the Process as a permanent procedure. Subsequently, any permanent change he may make in the approved process must be approved in writing by the Government. Testing for the adequacy of the approved procedure shall be in accordance with 4.2.4.1(a).

(a) Mine. The mine shall be contaminated with the liquid VX and the spot marked. The mine shall be processed through the decontamination section of the filling line (4.2.4.1) and then tested for agent contamination as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:28Z
Check the source to verify that this is the current version before use.

The ticket (4.2.4.1(c)) is removed from its envelope and the sensitive surface of the ticket wetted with two drops of buffer solution (4.2.4.1(e)). The ticket is replaced in the envelope and the buffer solution kneaded into the ticket for approximately 15 seconds. The ticket is removed from the envelope and the sensitive surface of the ticket is applied to the surface being checked for possible contamination. Sufficient finger pressure shall be applied to assure that firm contact of the entire sensitive surface area of the ticket is made with the surface of the mine being checked. The ticket shall remain in contact with the surface being checked for at least 15 seconds. The ticket is removed from the envelope and two drops of substrate (4.2.4.1(d) are added to the ticket. The ticket is replaced in the envelope and kneaded. within go seconds after the addition of substrate, the ticket should become blue if the area has been properly decontaminated.

(b) Closure leakage and contamination. The filled and closed mine shall be placed on its side in such a position that the fillport is completely covered by liquid and maintained in such position for a minimum period of 4 hours. Immediately prior to test, the mine shall be repositioned so that the fillport is up. Five milliliters (ml) of buffer solution (4.2.4.1(e)) shall be poured into the area around the fillport to cover the closure. The solution shall be thoroughly stirred with a clean glass or polyethylene dropper calibrated with 0.5 ml increments. Withdraw from the closure well 0.5 ml of buffer solution with the dropper and discharge into a clean paper cup of suitable size which contains 9.5 ml of buffer solution. Stir the solution in the cup with the dropper tip. The ticket (4.2.4.1(c)) shall be removed from the envelope and is immersed so that the sensitive area is below the surface of the buffer solution and shall be held for 10 seconds without movement. The ticket shall be removed and placed in the envelope where it shall be kneaded for 15 seconds. The ticket shall be removed from the envelope and two drops of substrate (4.2.4.1(d)) shall be added to the ticket. The ticket shall be replaced in the envelope and again kneaded, If no agent was present on the closure area of tile mine, the color of the ticket should turn blue within 60 seconds. If the color of the ticket is not blue, the ticket shall be replaced in the envelope and kneaded for an additional 30 seconds. If the ticket does not turn blue this time, the mine is contaminated.

(c) Detector ticket. The ticket shall be in accordance with MIL-D-51083.

(d) Substrate solution. The substrate solution shall be in accordance with MIL-S-51189.

(e) Buffer solution. The buffer solution shall be made up by dissolving 6.07 grams of tris (hydroxymethyl) amino methane in 250" ml of distilled water; add 287 ml of 0.1N hydrochloric acid and dilute to 1 liter with distilled water. The pH shall shall be 8.0 ± 0.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:28Z
Check the source to verify that this is the current version before use.

4.2.4.2 Helium leakage. Prior to closing, the void above the agent shall be evacuated to a Pressure between 6.5 and 7.5 pounds per square inch absolute (psia) and filled with helium to a maximum pressure of 18.5 psia to insure a partial pressure due to helium of not less than 11 pounds per square inch (psi) after closing. The closure area shall be sealed off by a test head manifold to a vacuum system and a mess spectrometer type helium detector, The test head shall be evacuated to and maintained at a pressure lower than 20 microns of mercury during test. The leak detector shall be calibrated to respond to a leak of 1 x 10^{-6} atmospheric cubic centimeters per second of helium at the test head when a standard leak is introduced into the system at that point.

4.2.4.3 Filling level. The filling level shall be determined radiographically.

5. PREPARATION FOR DELIVERY

5.7 Packing, level A.

5.1.1 Containerized land mines, Three fuzes and three activators packaged as shown on Drawings F7549159 and F7548391 respectively shall be paced with three mines in a steel shipping drum as shown on Drawing D37-1-120.

5.1.2 Palletization. Thirty-six mines (12 drums), packed in accordance with 5.1.1 shall be palletized as shown on Drawing D37-1-126.

5.2 Marking. In addition to any special marking required in the contract or order, shipping containers and palletized loads shall be marked in accordance with Drawing D37-1-115.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number and date of this specification.

6.2 Filling and closing. When filled as specified in 3.2.1, the mine carries approximately 315 to 342 cubic inches of VX. It has been found that the burster tube seats properly when tightened to a torque of 150 foot pounds. This information is furnished as an aid to procurement and not as a criterion for filling and closing.

6.3 Precautions. Because of the hazards attending the filling of mines with VX and loading with explosives, suppliers are advised to exercise all necessary precautions to protect personnel from injury during filling and loading. Care should be taken to prevent wetting the closure thereby making void the results of the helium tests.

Source: https://assist.dla.mil -- Downloaded: 202610-22T07:28Z
Check the source to verify that this is the current version before use.

6.4 Mine decontamination. The following method of decontamination has been found to be satisfactory. This procedure is furnished with the distinct understanding that the supplier is not required to follow the method; that the provisions of the specification must be met regardless of the method of decontamination, and that the Government makes no warranty that mines decontaminated in accordance with the method will meet the specifications requirements.

(a) Subject the closure to a jet of steam at a minimum pressure of 100 pounds per square inch gage (psig) for 20 to 30 seconds.

(b) Subject the mine to a spray containing surface active agent (see 6.5) for a minimum period of 40 seconds at a minimum pressure of 20 psig in such a manner as to completely wash all surfaces. The munition shall then be rinsed with water to remove all traces of the surface-active solution. The temperature of the spray and rinse water shall be maintained at a range of 140° to 160° F.

6.5 Surface active agent. Nacconal available from Allied Chemical Dye Corporation and Turco Aviation Detergent are two agents which have been found adequate for the purpose of decontamination.

6.6 Buffer solution. The tris (hydroxymethyl) amino methane may be procures from sigma Chemical Company, St, Louis, Missouri.

6.7 Due to the hazardous nature of the VX agent and to assure a satisfactory test, the following precautions should be taken.

(a) Protective gloves should be worn during test,

(b) When the ticket is applied to the surface, the rubbing should not be such as to disintegrate ticket.

(c) Fresh rubber finger protectors should be used over the glove for each test to avoid contamination.

(d) Envelope should be opened by trimming only enough to remove ticket leaving the envelope suitable for kneading operations.

6.8 The following specifications reflect the components required to complete this item:

MIL-W-46985 - Mine, Chemical Agent, VX, M23; Inert Components.
MIL-F-11245 - Fuze, Mine, T17E2.
MIL-B-11507 - Booster, M120.
MIL-A-20515 - Activator, Ml.
MIL-B-60089 - Burster, M38; Initiator; M48 and Booster.

6.9 MIL-STD-105 may be used when approved by the procuring activity.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 1345-A199

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:28Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:28Z
Check the source to verify that this is the current version before use.

TO DETACH FORM – CUT ON THIS LINE

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS. This sheet is to be filled out by personnel (either Government or contractor) involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced documents, or to serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 JAN 54 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T07:28Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
ATTN: SMUEA-QAES
Edgewood Arsenal, Maryland 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:28Z
Check the source to verify that this is the current version before use.